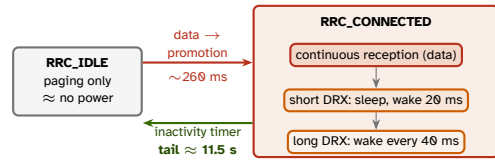


In one line: On cellular a small request is dominated not by bytes but by **waking the radio** (promotion from idle: $\sim 0.1\text{--}0.3\text{ s}$ on LTE, up to 2 s on 3G, then a **tail** of seconds at high power) and by **round trips**: DNS + TCP + TLS before the first byte, each one cellular RTT. So: **fewer, earlier, bigger** requests over connections you **already have** — and measure the phases, not the total.

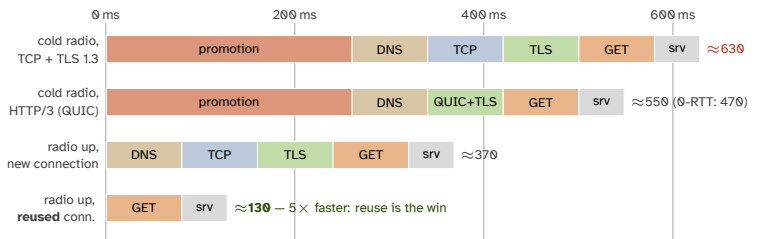
LTE radio (RRC) states



Measured on one LTE network (Huang et al., MobiSys 2012): promotion 260 ms, tail 11.576 s; on 3G, the radio stays up $\approx 11.5\text{ s}$ (tail) whatever you do — so **piggyback**: send queued analytics and prefetch *now*, while the tail is paid for.

3G: IDLE $\rightarrow$ DCH up to 2 s , FACH $\rightarrow$ DCH up to 1.5 s (HPBN). 5G NR adds **RRC_INACTIVE**: HPBN, active connection: 2G $300\text{--}1000\text{ ms}$ · 3G $100\text{--}500\text{ ms}$ · 4G $< 100\text{ ms}$ RTT. A 1 MB body adds several RTTs of TCP slow start (initial window ≈ 10 segments).

One small GET, RTT 80 ms — drawn to scale



How it works

- **Radio states (RRC):** the phone holds a dedicated channel only in CONNECTED. Getting there is a signalling exchange (**promotion**); leaving is on an **inactivity timer** set by the carrier, and the radio burns near-peak power during that **tail**. Inside CONNECTED, DRX sleeps between short wake-ups.
- **Consequence:** a 1 KB ping every 30 s keeps the radio up $\sim 40\%$ of the time (a tail each); every 10 s, always. Sent together they pay *one* promotion + *one* tail.
- **Round trips before a byte:** DNS (1, unless cached) + TCP (1) + TLS 1.3 (1; TLS 1.2: 2) + request (1) — 4 RTTs. QUIC merges transport + TLS (3); a reused connection needs 1. Big bodies then pay **slow start**: the window doubles per RTT from $\approx 14\text{ KB}$ (RFC 6928).
- **Loss** on cellular is common: with HTTP/2 over TCP one lost packet stalls every stream; HTTP/3 stalls only its own stream and **migrates** across Wi-Fi $\leftrightarrow$ cellular.

Connection reuse — the biggest lever

- **One shared URLSession** per configuration: its pool holds the warm connections. A session per request = full handshake every time.
- **HTTP/2:** one connection, many streams — no sharding. **Coalescing** (RFC 9113 §9.1.1): a connection may serve another host if its certificate covers that host (browsers also want the same IP); the server refuses with **421 Misdirected Request**. Put `api.` and `img.` on one cert and one IP.
- **Idle pool timeout is not API:** Apple DTS — typically 30 s on HTTP/1.1, longer on HTTP/2, **a few seconds on WWAN**. “Warm” after a pause often means a new handshake.

Example — where did the time go?

```
func urlSession(_ s: URLSession, task: URLSessionTask,
               didFinishCollecting m: URLSessionTaskMetrics) {
    for t in m.transactionMetrics where t.resourceFetchType == .networkLoad {
        func ms(_ a: Date?, _ b: Date?) -> Int { // -1 = phase skipped
            guard let a, let b else { return -1 }
            return Int(b.timeIntervalSince(a) * 1000)
        }
        log("dns", ms(t.domainLookupStartDate, t.domainLookupEndDate),
            "tcp", ms(t.connectStartDate, t.secureConnectionStartDate),
            "tls", ms(t.secureConnectionStartDate, t.secureConnectionEndDate),
            "ttfb", ms(t.requestStartDate, t.responseStartDate),
            "reused", t.isReusedConnection, // iOS 10
            "cell", t.isCellular, t.networkProtocolName ?? "?" // iOS 13
        )
    }
}
```

Fewer, earlier, bigger

- **Kill waterfalls:** 5 dependent calls = 5+ RTTs. Parallelise what is independent; a screen-level endpoint / BFF or GraphQL for what is not.
- **Prefetch** what the next screen almost surely needs, while the radio is already up; **batch** analytics and flush on a user-triggered request.
- **Priorities:** `URLSessionTask.priority` (0–1, default 0.5) — a hint; **cancel** off-screen image loads; defer bulk work to a background session.
- **Warm up early:** start the first API call (its DNS + TLS) at launch, in parallel with UI setup.

Adapt to the network

`NWPathMonitor`: `isExpensive` (cellular, hotspot), `isConstrained` = **Low Data Mode** (iOS 13) $\rightarrow$ skip prefetch, smaller images; per request `allowsConstrainedNetworkAccess`. `waitsForConnectivity` (iOS 11) waits for a path instead of failing. `timeoutIntervalForRequest` (60 s) = max *idle gap*; `timeoutIntervalForResource` (7 days) = whole task.

Measure

Metrics per phase (example) from the field · **Instruments** $\rightarrow$ **Network** $\rightarrow$ **HTTP Traffic** (Xcode 13): sessions $\rightarrow$ tasks $\rightarrow$ transactions, on device · **Network Link Conditioner** (Settings $\rightarrow$ Developer): 3G, LTE, *Very Bad Network*, 100% loss.

Traps

- ! “Fast on office Wi-Fi” — test on Link Conditioner and a real cell.
- ! “Need more bandwidth” — API calls are latency-bound: count RTTs.
- ! A new URLSession per request — a handshake every time.
- ! Polling every 10 s — the radio never leaves the tail. Push or batch.
- ! `timeoutIntervalForRequest` resets on every packet: not a total.

Remember

“Wake once, round-trip less, reuse always, measure the phases.”